Windows Networking Services

- **DNS in Windows Server Theory :Introduction to DNS: Concepts, zones, and records. Configuring DNS in Windows Server. Practical :Set up DNS zones and create records. Task: Simulate a DNS resolution scenario and submit findings.**

1. Open Server Manager on your Windows Server VM, add the DNS Server role, and take a screenshot showing the DNS role installed.

ANS:

Step 1: Open Server Manager

1. Log in to your Windows Server.
2. Open Server Manager.
3. Click on Manage in the top-right corner.
4. Select Add Roles and Features.

Step 2: Select Installation Type

1. Click Next.
2. Select Role-based or feature-based installation.
3. Click Next.

Step 3: Select Server

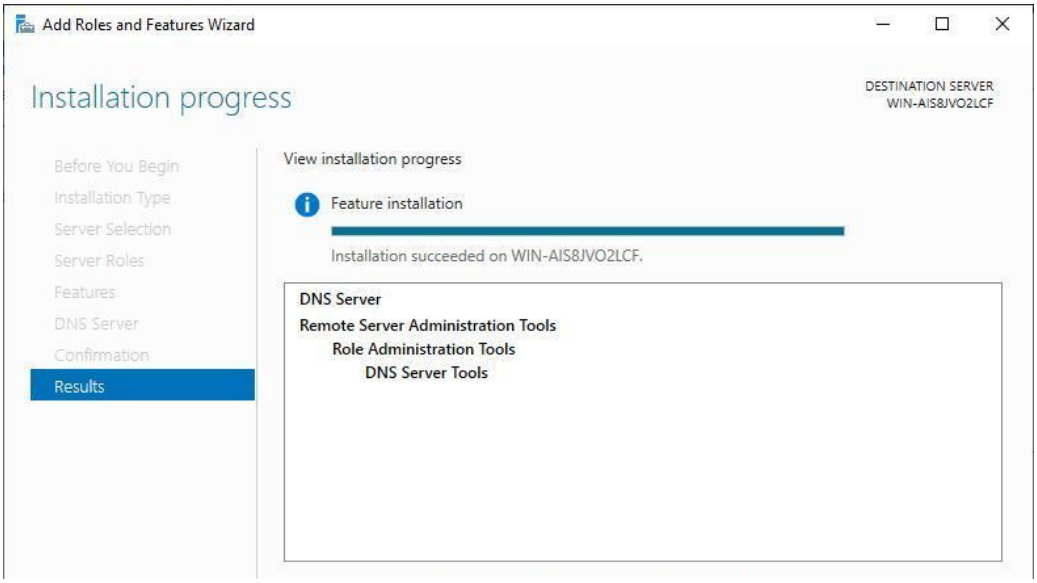
1. Select your local server from the Server Pool.
2. Click Next.

Step 4: Select DNS Server Role

1. Under Server Roles, locate DNS Server.
2. Check the DNS Server option.
3. Click Add Features when prompted.
4. Click Next.

Step 5: Install DNS Server

1. Continue clicking Next.
2. Click Install.
3. Wait until the installation completes.
4. Click Close.


Add Roles and Features Wizard
Installation progress
DESTINATION SERVER
WIN-AIS8JVO2LCF
Before You Begin
Installation Type
Server Selection
Server Roles
Features
DNS Server
Confirmation
Results
View installation progress
Feature installation
Installation succeeded on WIN-AIS8JVO2LCF.
DNS Server
Remote Server Administration Tools
Role Administration Tools
DNS Server Tools

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2. In DNS Manager, create a new Forward Lookup Zone named foodies.local and add an A record for zomato.foodies.local pointing to 192.168.1.50.

ANS:

Step 1: Open DNS Manager

1. Open Server Manager.
2. Click Tools.
3. Select DNS.
4. DNS Manager will open.

Step 2: Create Forward Lookup Zone

1. Expand your server name.
2. Right-click Forward Lookup Zones.
3. Select New Zone.
4. Click Next.
5. Select Primary Zone.
6. Click Next.
7. Enter the Zone Name: foodies.local
8. Click Next.
9. Select Create a new file with this file name.
10. Keep the default filename.
11. Click Next.
12. Select Do not allow dynamic updates.
13. Click Next and Finish.

Your Forward Lookup Zone should now appear as:

```
DNS
└── Server Name
    └── Forward Lookup Zones
        └── foodies.local
```

Step 3: Create A Record

1. Expand Forward Lookup Zones.
2. Right-click foodies.local.
3. Select New Host (A or AAAA).
4. Enter the following details:

New Host configuration

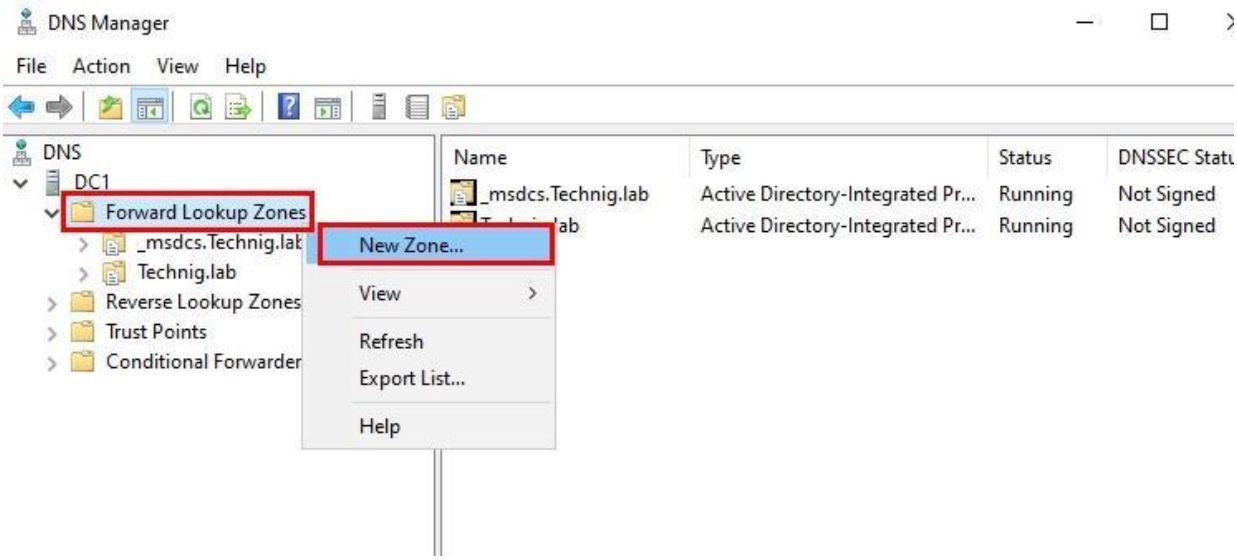
Field	Value
Name	zomato
Fully Qualified Domain Name	zomato.foodies.local
IP Address	192.168.1.50
Create associated PTR record	Unchecked

Expected configuration

Click Add Host, then click OK and Done.

Your DNS record should look like this:

Name	Type	Data
zomato	Host (A)	192.168.1.50



**3. Add a CNAME record in the foodies.local zone so that swiggy.foodies.local points to zomato.foodies.local.

Hint: Use the 'New Alias (CNAME)' option and test both names with nslookup.**

ANS:

Step 1: Open New Alias (CNAME)

1. Open DNS Manager.
2. Expand Forward Lookup Zones.
3. Right-click foodies.local.
4. Select New Alias (CNAME).

Step 2: Configure CNAME Record

Enter the following details:

New Resource Record configuration

Field	**Value**
Alias Name	swiggy
Fully Qualified Domain Name (FQDN)	swiggy.foodies.local
Fully Qualified Domain Name for Target Host	zomato.foodies.local

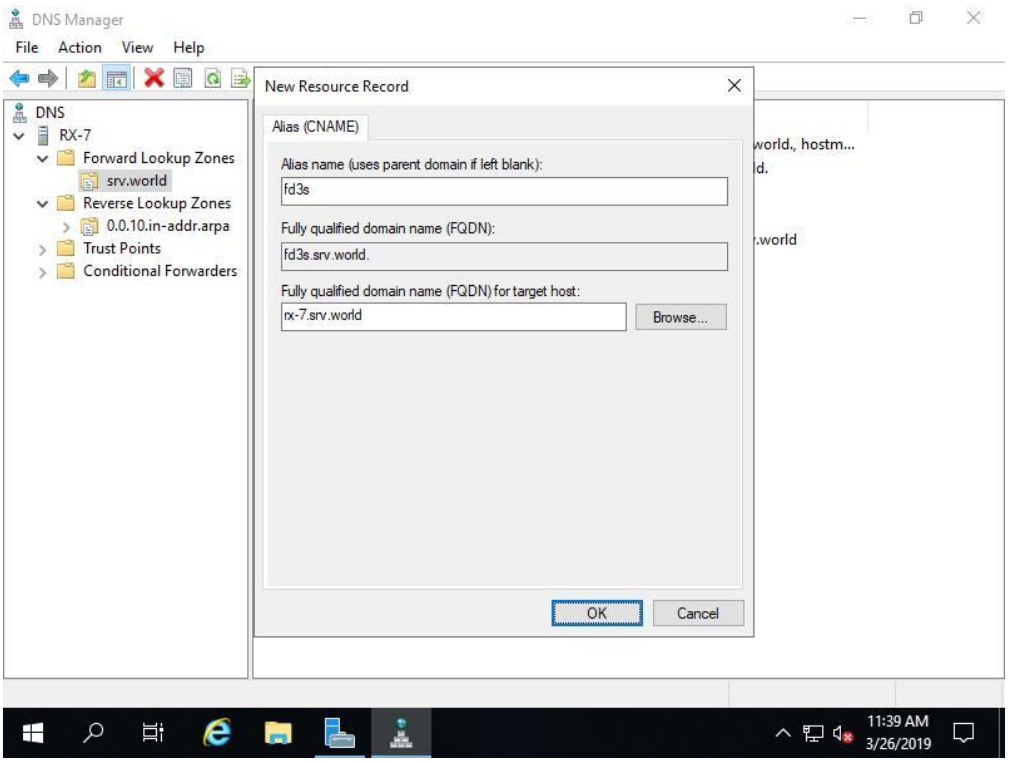
Expected configuration

Click OK to create the record.

Step 3: Verify DNS Records

Your DNS Manager should now contain the following records:

Name	**Type**	**Value**
zomato	Host (A)	192.168.1.50
swiggy	Alias (CNAME)	zomato.foodies.local



4. Simulate a DNS resolution: On a client machine, use nslookup to resolve swiggy.foodies.local and zomato.foodies.local, then explain what happens and why in 3-4 lines.

ANS:

Step 1: Configure Client DNS Settings

On your Windows 10 client:

1. Open Control Panel.
2. Navigate to Network and Internet → Network Connections.
3. Right-click your Ethernet adapter and select Properties.
4. Select Internet Protocol Version 4 (TCP/IPv4).
5. Click Properties.
6. Select Use the following DNS server addresses.

Enter your DNS server's actual IP address.

Example:

Field	**Value**
Preferred DNS Server	Your Windows Server IP
Alternate DNS Server	Leave blank

Important: The DNS server IP must be reachable from the client. The example A record IP,

192.168.1.50, is the destination address and does not automatically mean it is your DNS server's IP.

```
PS C:\Users\akbar> nslookup akbar.local
Server:  serv.akbar.local
Address:  192.168.10.2

Name:    akbar.local
Address:  192.168.10.2

PS C:\Users\akbar> nslookup 192.168.10.2
Server:  serv.akbar.local
Address:  192.168.10.2

Name:    akbar.local
Address:  192.168.10.2
```

Step 2: Open Command Prompt

On your Windows client:

1. Press Windows + R.
2. Type cmd.
3. Press Enter.

First, clear the DNS cache: ipconfig /flushdns

Step 3: Test Zomato DNS Resolution

Execute:

nslookup zomato.foodies.local

Expected output (example):

Server: server.foodies.local

Address: 192.168.1.10

Name: zomato.foodies.local

Address: 192.168.1.50

Step 4: Test Swiggy DNS Resolution

Execute:

nslookup swiggy.foodies.local

Expected output (example):

Server: server.foodies.local

Address: 192.168.1.10

swiggy.foodies.local canonical name = zomato.foodies.local

Name: zomato.foodies.local

Address: 192.168.1.50

Note: The DNS server name and IP in these examples are illustrative. Your actual output will depend on your server configuration.

Step 5: Explanation (3–4 Lines)

Write this explanation in your assignment:

When we execute nslookup zomato.foodies.local, the DNS server finds its A record and returns the IP address 192.168.1.50. When we execute nslookup swiggy.foodies.local, the DNS server finds its CNAME record and redirects the lookup to zomato.foodies.local. The DNS server then resolves the target hostname using its A record and returns 192.168.1.50. Therefore, both domain names resolve to the same IP address.